

# A Connectivity-Driven Multi-Scale Atlas of Human Deep White Matter Junctions

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## Abstract

Human white matter is usually mapped as a set of named pathways, yet many of its most important territories are junctions where multiple systems converge, cross, run in parallel, and diverge. To map these regions directly, we developed a connectivity-driven, multi-scale atlas of human deep white matter junctions. Each white matter voxel was assigned a connectivity signature based on its tractography-derived endpoint connections and described by its combination of association, commissural, projection, and cerebellar pathway components. We then combined these voxel-wise maps from 105 healthy adults using intensity-driven joint label fusion to produce a population atlas of 281 labels. The resulting White Matter Junction Atlas organizes these labels across several anatomical scales. At the broadest level, the labels collapse into four major pathway systems - association, commissural, projection, and cerebellar - and recover the classical concentric organization of cerebral white matter, with projection and commissural pathways forming a deep core and association pathways occupying a more external zone. At intermediate scales, the atlas identifies nine inter-system *junction* classes and 31 overlap or *bottleneck* classes, capturing territories where distinct or repeated pathway components share the same anatomical volume. Scan-rescan analysis in 41 subjects showed scale-dependent reproducibility, with highest agreement for broad system classes and lower agreement for fine labels. The atlas complements named tracts and classical regional atlases by providing a common anatomical reference for localizing white matter findings, studying tractography uncertainty, and investigating lesions, development, aging, and surgical risk.

## Introduction

Understanding the complex architecture of the human brain requires standardized, broadly applicable maps. Macroscale neuroscience addresses this challenge by describing the brain at the level of anatomical regions, long-range pathways, and distributed systems that can be consistently identified across individuals. Since the early 20<sup>th</sup> century, a central objective in macroscale neuroscience has been the delineation of the cerebral cortex into distinct areas. Classical studies of cytoarchitecture, myeloarchitecture, and chemoarchitecture divided the cortical sheet into regions whose structure, function, and topography change at recognizable borders (Brodman, 1909; Campbell, 1905; Felleman & Van Essen, 1991; Flechsig, 1920; Smith, 1907; Vogt, 1910, 1911; Zilles, Schleicher, Palomero-Gallagher, & Amunts, 2002), with modern multimodal neuroimaging extending this work by enabling robust, noninvasive identification of these regions in living individuals (Amunts, Mohlberg, Bludau, & Zilles, 2020; Glasser et al., 2016). Beyond reducing complexity, such maps provide a standardized nomenclature that creates common ground across studies and across subjects. Much like “cities” on a map, these cortical and subcortical regions serve as the fundamental nodes for structural and functional analyses.

In parallel, substantial progress has been made in mapping the “roads” between these cities, the white matter pathways of the brain. Comprising approximately 160,000 kilometers of myelinated axons (Marnier, Nyengaard, Tang, & Pakkenberg, 2003), white matter forms the structural infrastructure linking distributed cortical and subcortical regions (Thiebaut de Schotten & Forkel, 2022). Advances in diffusion MRI and fiber tractography (Dell’Acqua, Descoteaux, & Leemans, 2024) have made it possible to delineate these pathways in vivo, allowing researchers to segment and study large-scale association, projection, and commissural bundles. Yet while the field now has increasingly robust maps of major pathways, it still lacks a comprehensive map of the complex intersections, interchanges, and bottleneck-like territories where these pathways merge, cross, and diverge (Maier-Hein et al., 2017).

White matter atlases have evolved alongside the methods used to generate them. Early voxel-based templates assigned a single anatomical label to each location in standard space (S. Mori, Oishi, & Faria, 2009; S. Mori et al., 2008; Oishi et al., 2008). Later tractography-based approaches shifted the emphasis from location to tracts, producing volumetric and streamline-based atlases of named fasciculi and enabling automated segmentation of major pathways across subjects (M. Catani & Thiebaut de Schotten, 2008; Hansen et al., 2020; Radwan et al., 2022; Yeh et al., 2018; Zhang et al., 2018). More recent frameworks have moved closer to the true complexity of white matter by allowing multiple labels within a voxel or by mapping pockets of crossing fibers (Qi & Arfanakis, 2021; Volz, Cieslak, & Grafton, 2018). Even so, these approaches still primarily describe pathways, streamline occupancy, or local orientation structure. They do not provide a whole-brain mapping and classification of the junction areas formed where pathways meet.

Those meeting points are a recurring feature of deep white matter organization. Commissural fibers crossing the midline encounter ascending and descending projection systems in the corona radiata, while long association pathways sweep anterior-posterior through many of the same territories. Some of these interactions appear as classical “crossing fiber” regions, where distinct fiber populations retain different local orientations (Alexander & Seunarine, 2010; Jeurissen, Leemans, Tournier, Jones, & Sijbers, 2013; Tuch et al., 2002). Others form bottleneck-like configurations, where multiple pathways are funneled through a narrow space and may briefly

share the same orientation before separating again (Maier-Hein et al., 2017; Schilling et al., 2021). In the road metaphor, these regions are the interchanges or intersections of white matter organization. For tractography, these regions might represent traffic jams, where they present a major computational challenge because local diffusion information may be insufficient to determine which pathway should continue where. They also present a broader anatomical problem. When a result falls within one of these crowded deep white matter territories, current atlases often provide no clear way to describe what that region is, which systems pass through it, or how it fits within the larger organization of the brain.

Here, we introduce a connectivity-driven, multi-scale atlas of deep human white matter topography centered on these junction areas. We define *junction* as a geometric descriptor referring to a discrete volume of white matter where multiple distinct fiber systems occupy the same spatial coordinates. The term junction accurately characterizes these regions as sites of structural convergence, where independent pathways interweave or 'kiss' while maintaining their individual connectivity profiles. Using tractography-derived connectivity signatures and population fusion across subjects, we parcellate white matter into 281 reproducible labels organized within a hierarchical framework. At the broadest level, the atlas recovers the major pathway systems of the brain and their concentric organization. At finer levels, it identifies inter-system junctions, bottleneck-like overlap patterns, and spatially specific territories with distinct connectivity profiles. The resulting atlas provides a standardized nomenclature, population probability maps, and a framework for locating deep white matter findings within a common anatomical reference. In doing so, it extends white matter mapping beyond named pathways alone and toward the organization of the regions where those pathways meet, interact, and converge.

## Results

### Overview of the white matter junction atlas

We generated a multi-scale atlas of human deep white matter topography composed of 281 reproducible labels. **Figure 1** visualizes the volumetric atlas in standard space and shows that these labels tile the deep white matter with clear boundaries and marked variation in size and shape. Some labels occupy broad and relatively continuous territories, whereas others are smaller and more spatially restricted, particularly around classical pathway-rich zones such as the centrum semiovale, corona radiata, and internal capsule. Across hemispheres, the atlas shows a largely symmetric organization, while still preserving regional variability in the density and extent of labeled territories.

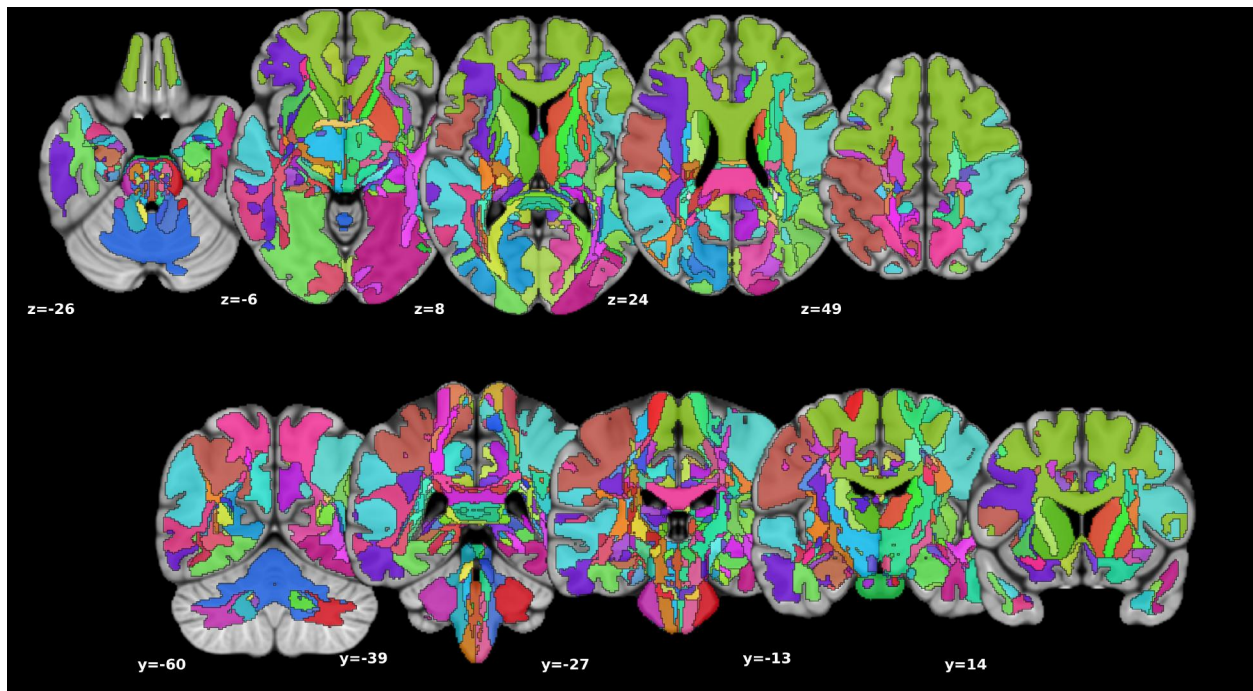
The fundamental unit of this atlas is the *connectivity signature* assigned to each white matter voxel. As illustrated in **Figure 2**, streamline endpoints were first assigned to 15 coarse anatomical regions spanning cortical, subcortical, cerebellar, and brainstem regions (Figure 2A). For each white matter voxel (Figure 2B), all streamlines passing through that location were identified and converted into a voxel-specific connectivity profile ( $15 \times 15$  matrix) (Figure 2C) based on their endpoints – which takes the form of association, projection (and cerebellar), and commissural pathways (Figure 2D). The upper triangular portion of this matrix (including within-regions connections) was unraveled into a 120-element vector and binarized using a 10% threshold (Figure 2E), so that only connections accounting for more than 10% of streamlines through that

voxel contributed to the signature. Subject-level signature-label maps were then fused across 105 HCP participants using joint label fusion (Figure 2F), with each subject's FA image used as the guidance image and the corresponding signature-label map used as the atlas label image. This produced a population consensus atlas of 281 labels.

**Figure 3** highlights representative labels from the atlas and the connectivity signatures used to define them across individual subjects. Each label is associated with a binarized signature composed of inter-regional connections, with each nonzero component corresponding to a specific anatomical system type. Connections within the same hemisphere between cortical lobes are classified as association (A), connections between cortex and subcortical structures as projection (P), interhemispheric connections as commissural (C), and connections involving the cerebellum as cerebellar (Cb), while anatomically infeasible pairings are denoted X. These signatures therefore provide an interpretable summary of the dominant pathway systems represented within each label.

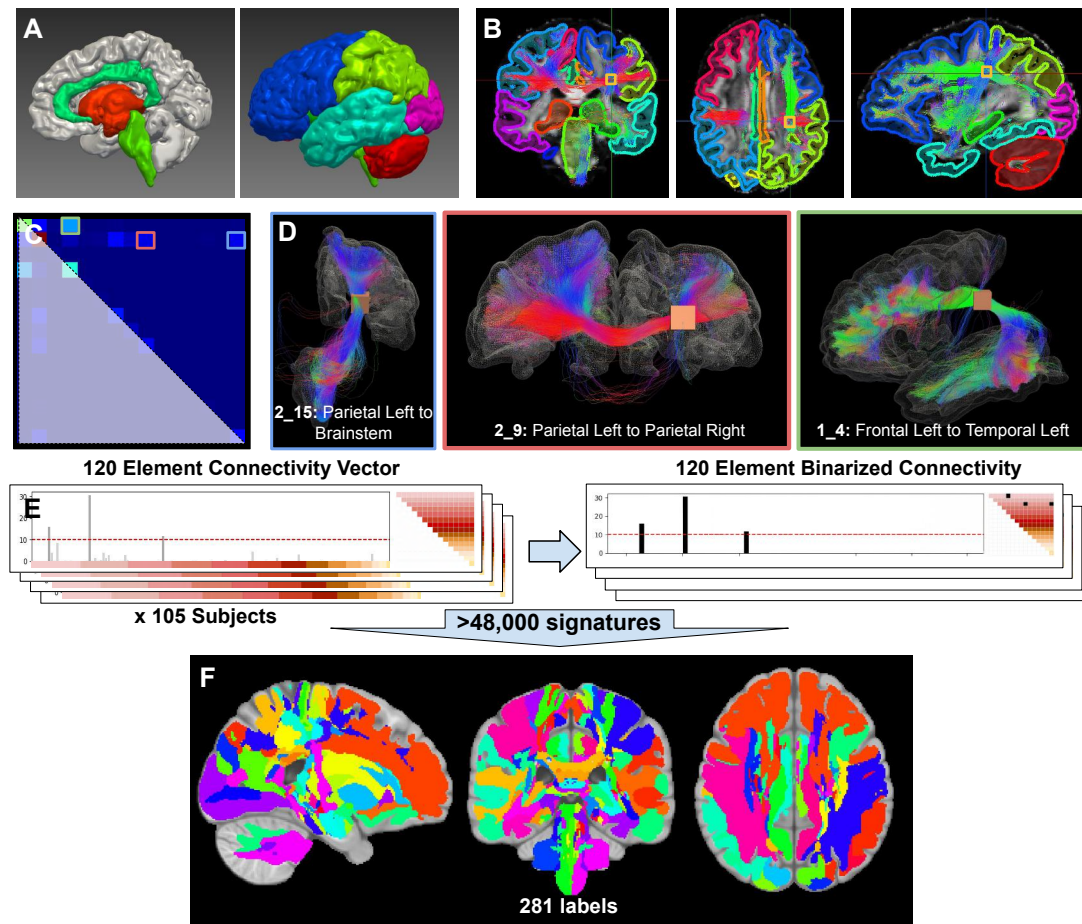
Across subjects, the broad spatial pattern of each label is preserved, although local differences in size, extent, connected components, and precise boundaries are evident. Several examples include multiple spatially separate components assigned the same label, yet the number of components and their spatial relationship to one another remain remarkably consistent across subjects.

To make this large set of labels neuroanatomically interpretable, we organized the atlas into a progressive multi-scale hierarchy. The following sections describe this hierarchy which captures the primary pathway systems of the brain and their relation to classical concentric white matter organization.

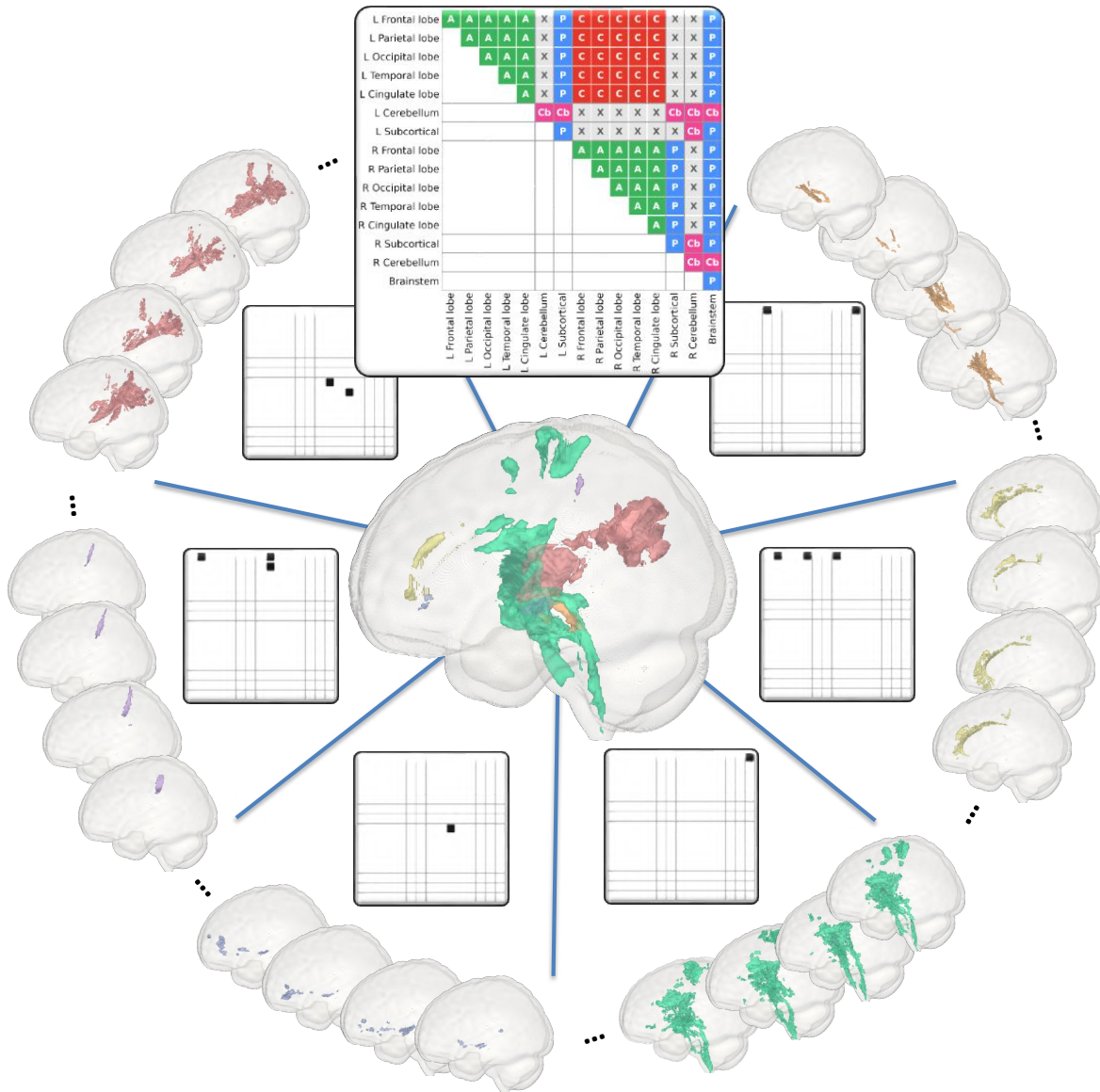


**Figure 1. White matter junction atlas in standard space.** Montage of axial slices (top) and coronal slices (bottom) showing the 281 atlas labels across the white matter. Slice locations are indicated in MNI coordinates (mm). Colors were assigned randomly for visualization and do not encode hierarchy or anatomical class. The montage illustrates the

spatial extent of the atlas, the sharp boundaries between neighboring territories, and the marked variation in label size and complexity across regions.



**Figure 2. Construction of the white matter junction atlas from voxel-wise connectivity signatures.** (A) The brain was first divided into 15 cortical, subcortical, cerebellar, and brainstem regions of interest using FreeSurfer-derived (Fischl, 2012) labels. (B) For a given white matter voxel, all streamlines passing through that location were selected. (C) These streamlines were then converted into a voxel-specific  $15 \times 15$  connectivity matrix based on their endpoints. (D) Individual connections contributing to that matrix can be visualized separately. (E) The upper triangular portion of the connectivity matrix was unraveled into a 120-element vector representing the proportion of streamlines assigned to each inter-regional connection, then binarized using a 10% threshold (to focus on dominant connections) to define the voxel's connectivity signature. (F) Joint label fusion across 105 HCP subjects grouped voxels with similar connectivity signatures, yielding 281 reproducible labels across the population.



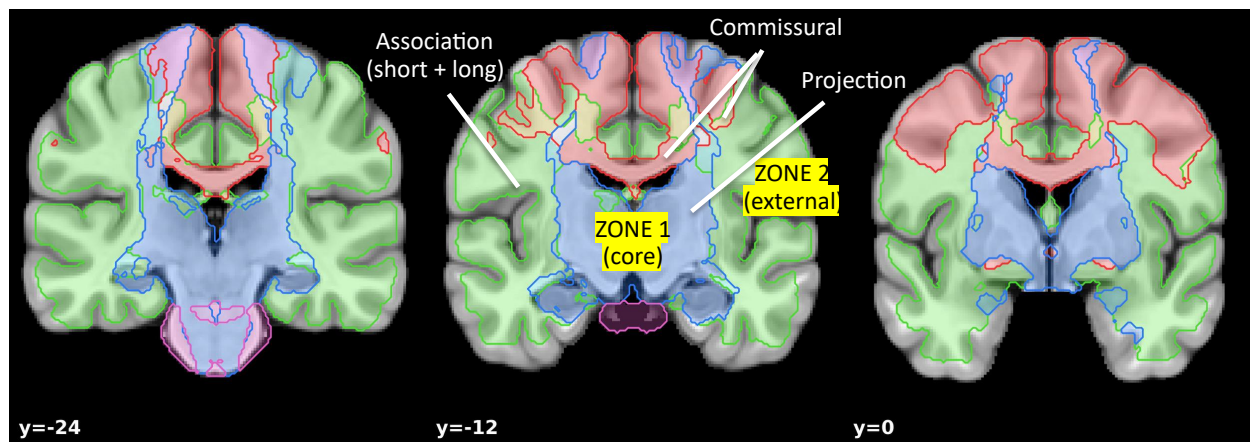
**Figure 3. Representative labels and their connectivity signatures across individual subjects.** Six example labels are shown in template space and in four randomly selected individuals to illustrate their spatial distribution, subject-level correspondence, and associated connectivity signatures. Matrices summarize the system-level interpretation of signature components, with association (A), projection (P), commissural (C), and cerebellar (Cb) connections derived from the underlying inter-regional edges; X denotes anatomically infeasible pairings. Several labels include multiple spatially separate connected components that are assigned the same label, reflecting a shared connectivity signature despite spatial discontinuity. Across subjects, the overall arrangement of these components is preserved, with local variation in size, extent, and boundary position.

#### Level 1: Primary systems (4 classes) and Meynert's concentric zones

At the broadest level of the hierarchy, the 281 labels can be grouped into four primary pathway systems: association, commissural, projection, and cerebellar. **Figure 4** shows the spatial distribution of these system-level classes across the deep white matter. Because individual atlas labels may contain more than one dominant system, these classes can overlap in space. Even at

this coarse level, the organization is highly structured. Association territories occupy much of the lateral and subcortical white matter beneath the cortex, commissural territories are centered along the midline and corpus callosum, projection territories course through the corona radiata and deeper periventricular white matter, and cerebellar territories are concentrated in brainstem and cerebellar connecting regions.

This large-scale partition recapitulates the classical concentric organization of cerebral white matter (Dannhoff, Poudel, Bhattarai, Kalthur, & Maldonado, 2023; Garcia, Wang, & Kroenke, 2021; Schüz, Braitenberg, & Miller, 2002) originally described by Meynert in 1885 (Marco Catani, 2025; Meynert, 1885). In coronal view, projection and commissural pathways define what might be called the “core” zone, forming the deep scaffold around the lateral ventricles, whereas association pathways occupy the “external” zone and extend more peripherally beneath the cortex. This pattern is consistent with the view that the core zone is dominated by early-forming long projection and commissural systems, while the external zone expands with association pathways and their terminal portions. This level establishes the broad anatomical scaffold on which the finer levels of the atlas are built.



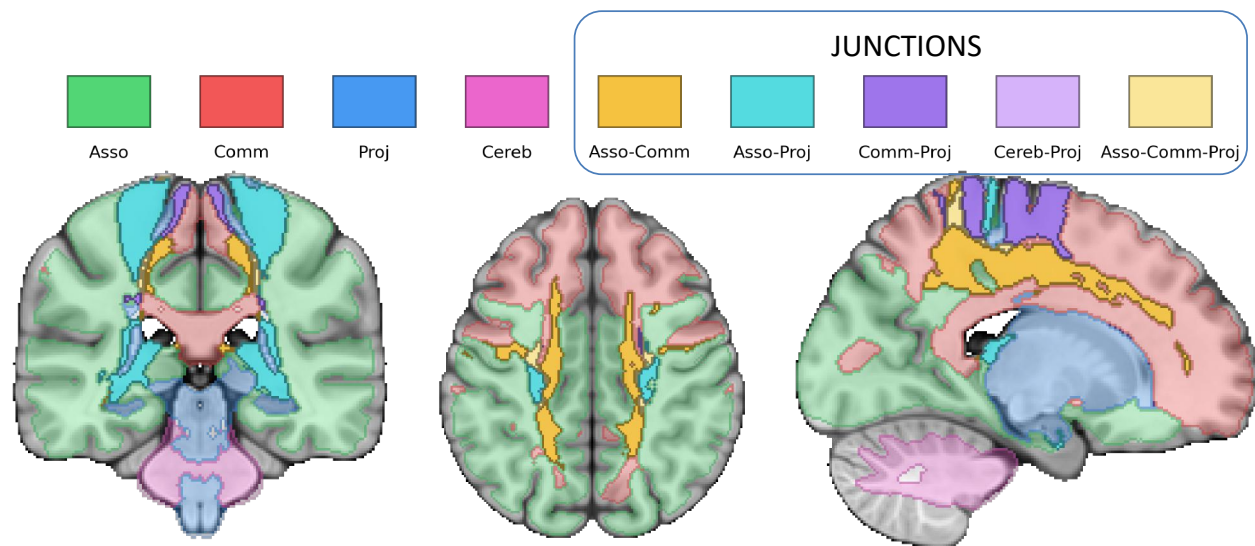
**Figure 4. Level 1 classification of the Junction Atlas into major white matter pathway systems.** Atlas labels are grouped into four major pathway systems: association (green), commissural (red), projection (blue), and cerebellar (pink). These classes can overlap in space because a single label may contain more than one dominant system. The resulting pattern recovers the classical concentric organization of cerebral white matter, with projection and commissural pathways forming the deep core zone and association pathways occupying the more peripheral external zone.

#### Level 2: Inter-system crossings and junctions (9 classes)

The second level of the hierarchy groups atlas labels into nine non-overlapping system-combination classes defined by the presence of one or more major pathway systems within a territory. In addition to the four single-system classes association, commissural, projection, and cerebellar, this level identifies five multi-system junction classes: association-commissural (Asso-Comm), association-projection (Asso-Proj), commissural-projection (Comm-Proj), cerebellar-projection (Cereb-Proj), and association-commissural-projection (Asso-Comm-Proj). **Figure 5** shows these nine classes highlighting the multi-system junction territories, with the corresponding

five junction classes also displayed in three dimensions. At this level, the atlas identifies spatially distinct territories where major pathway systems co-occur.

Several spatial motifs are immediately apparent. The most prominent is a blade-like junction territory in superior frontal and parietal white matter, dominated by association-commissural and association-commissural-projection configurations. A second major concentration is centered near the precentral and postcentral regions, where association-projection junctions form an extended band along the deep white matter beneath sensorimotor cortex. More focal commissural-projection and cerebellar-projection territories are also present in expected deep and infratentorial locations. Although these classes are defined by connectivity signature rather than by local fiber orientation, many of these territories coincide with regions recognized as sites of pathway crossing (i.e., crossing fiber “pockets”(Volz et al., 2018)), and also sheet-like structures – large, near-rectilinear grid of interwoven paths in the longitudinal and medio-lateral axes of the brain(Tax et al., 2016). **Figure 5** therefore captures the first anatomical level at which the atlas begins to isolate junctions themselves as reproducible white matter territories, rather than simply the major pathways that traverse them.



**Figure 5. Level 2 organization of the Junction Atlas into inter-system junction classes.** At the second level of the hierarchy, atlas labels are grouped into nine non-overlapping system-combination classes. The panels show the four single-system classes and five multi-system junction classes: association-commissural, association-projection, commissural-projection, cerebellar-projection, and association-commissural-projection. Prominent motifs include a blade-like association-commissural and association-commissural-projection territory in superior frontal and parietal white matter, and an extended association-projection junction region near the precentral and postcentral deep white matter.

### Level 3: Intra-system overlaps and bottleneck classes (31 classes)

The third level of the hierarchy resolves the atlas into 31 non-overlapping classes defined by pathway-system composition at a finer scale. Rather than simply recording which major systems are present, this level also distinguishes repeated components within a system, allowing territories to be separated into patterns such as A, A-A, A-A-P, or even C-C-C-C. These classes are often small and spatially restricted, but they are distributed throughout the deep white matter and provide a more precise description of where pathway overlap becomes especially dense or

internally complex. Practically, these classes mark locations where tractography is likely to be most ambiguous, because several candidate pathways may share the same local trajectory before separating again.

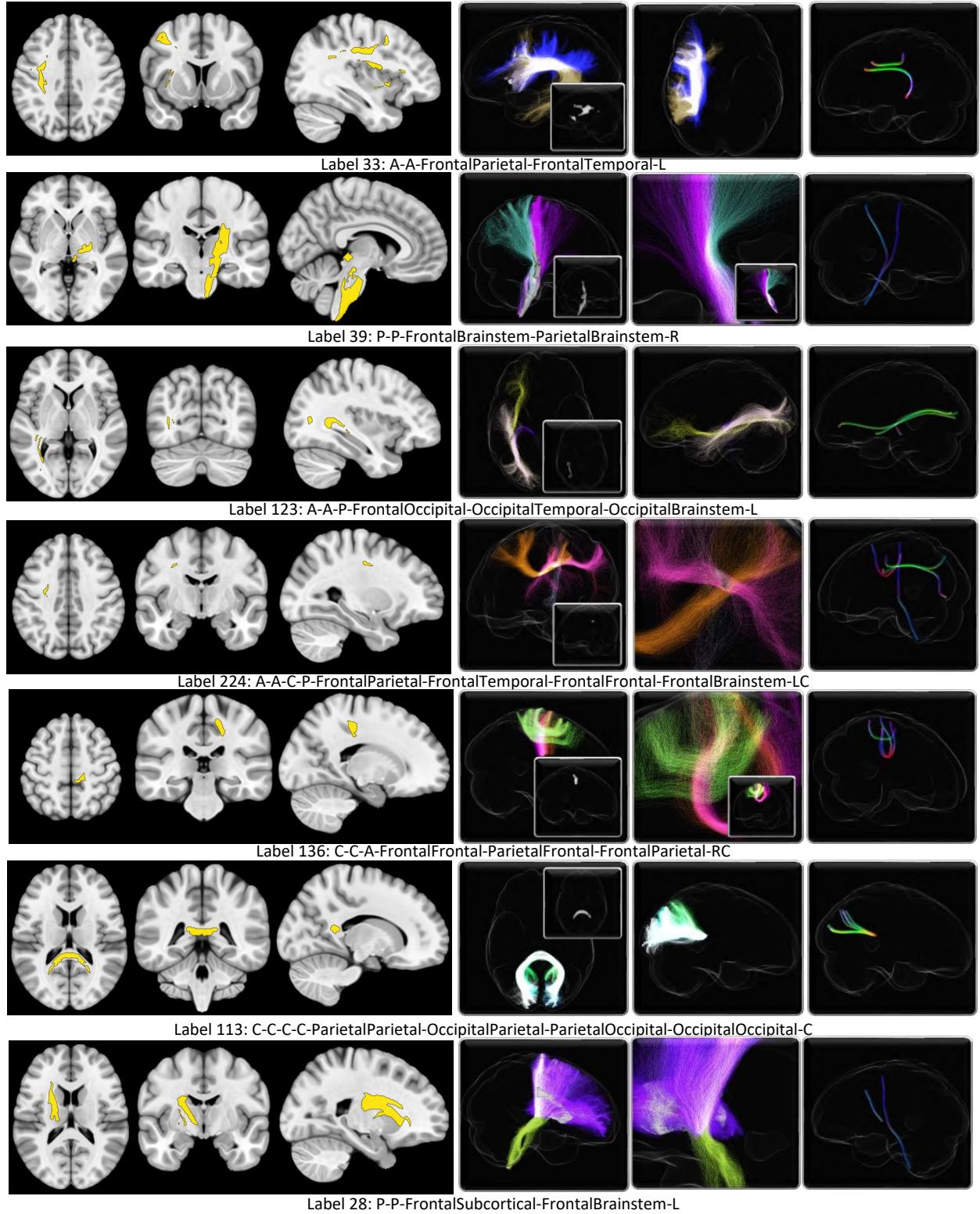
**Figure 6** illustrates representative examples of bottleneck-pattern classes. Some examples capture parallel overlap within a single major system. An association-association class in the superior fronto-parietal white matter, illustrated by Label 33 (A-A), occupies part of the superior longitudinal system (Mandonnet, Sarubbo, & Petit, 2018); it represents a high-density corridor where the long and anterior segments of the arcuate fasciculus are bundled together before fanning out toward their respective cortical targets. Similarly, a projection-projection example, Label 39 (P-P), delineates a vertical projection bottleneck near the transition from the fan-like corona radiata into the more compact internal capsule and descending brainstem pathways.

Other examples identify more heterogeneous confluence zones where disparate systems interweave. Label 123 (A-A-P) illustrates a posterior association-projection bottleneck in the left hemisphere, where long-range association fibers of the inferior fronto-occipital fasciculus (IFOF) and inferior longitudinal fasciculus (ILF) are funneled alongside posterior projection pathways. This region serves as a major "highway interchange" for visual and semantic information moving between the occipital lobe and the rest of the brain.

Label 224 (A-A-C-P) shows a more complex mixed-system configuration in the frontal lobe, combining association, commissural (corpus callosum) and projection components within a single reproducible atlas territory. These "multimodal" bottlenecks represent the most complex structural nodes in the deep white matter, where the spatial constraints of the brain force highly diverse functional streams into a shared anatomical volume.

Some Level 3 classes reflect dense intra-system bundling within commissural territories. Label 136 (C-C-A), for example, combines frontal and parietal commissural components with an association component. This creates a "tri-axial" junction where the callosal "bridge" is intersected by longitudinal "highways." Label 113 (C-C-C-C) identifies a commissural-rich bottleneck within the splenium of the corpus callosum. Here, the atlas resolves four distinct posterior commissural populations that, while sharing a similar medial-lateral orientation, maintain highly specific homotopic and heterotopic termination patterns. This example extends to the posterior callosum a principle demonstrated anatomically and tractographically in the frontal callosum (De Benedictis et al., 2016): callosal fibers can form heterogeneous homotopic and heterotopic populations and can become difficult to resolve when distinct populations run together or intermingle with other white matter systems. This represents the classic "bottleneck" challenge, even in territories with bundles largely in parallel, sharing similar trajectories within the same anatomical region.

By isolating these convergent parallel corridors, the third tier of the atlas extends white matter mapping beyond simple "crossing pockets" (Volz et al., 2018). It identifies the specific anatomical territories where the limits of dMRI resolution and tractography specificity are most likely to be tested, while also revealing recurring organizational motifs within the deep white matter.



**Figure 6. Level 3 organization of the junction atlas into bottleneck-pattern classes.** Representative examples of Level 3 classes are shown to illustrate the range of intra-system overlaps and bottleneck configurations captured by the atlas. Each panel includes the label identifier, its Level 3 class, and the specific connectivity pattern represented by that label. Insets show the spatial extent of the label as a 3D polygon, streamlines are colored by contributing bundle,

and the rightmost panel shows corresponding bundle centroids colored by local orientation. Examples include association-association bottlenecks, projection-projection bottlenecks, mixed association-association-projection configurations, and more complex association-association-commissural-projection and commissural-rich patterns. Bottlenecks vary in size, shape, and anatomical setting, ranging from broad parallel overlap zones to small but densely packed white matter interchanges.

### Additional granularities of the junction atlas

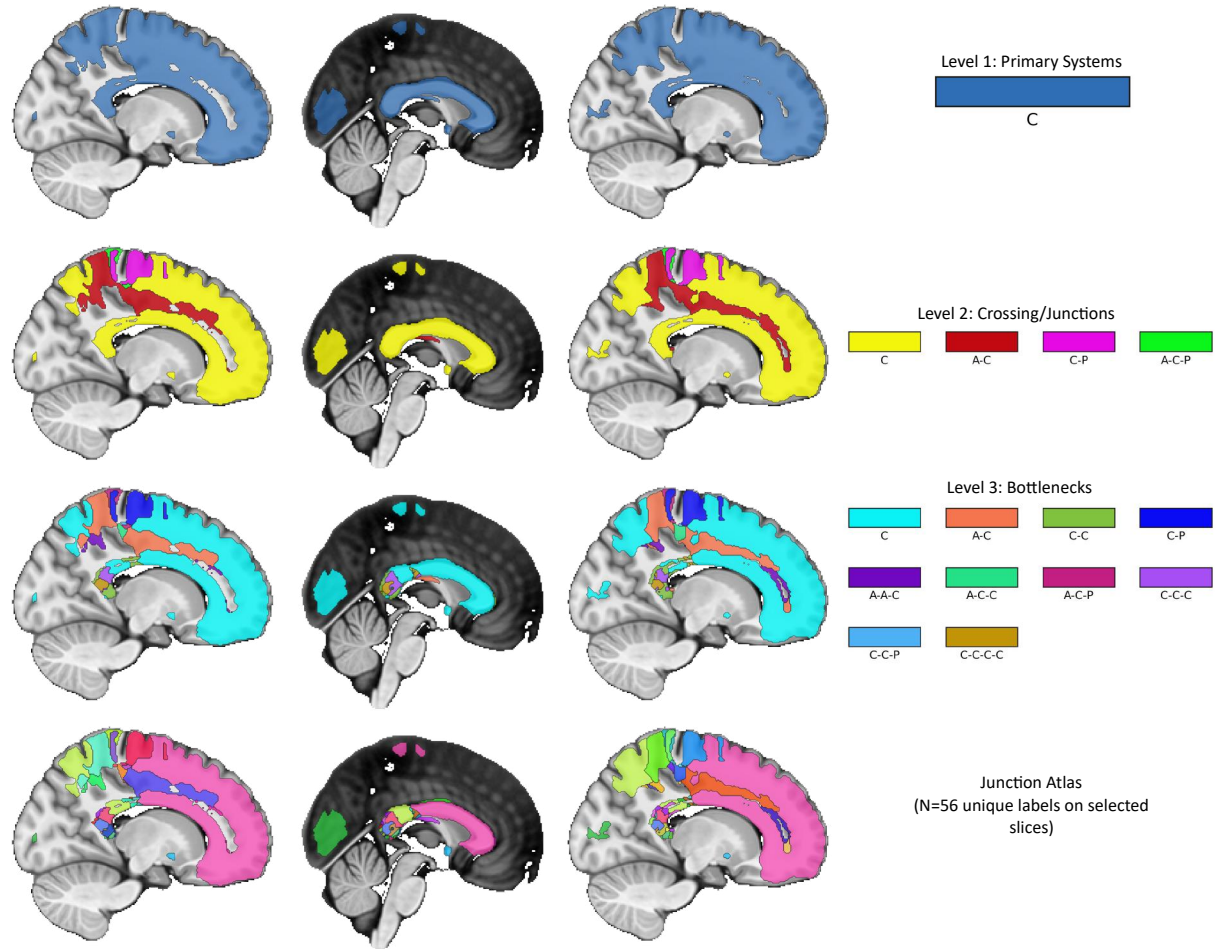
The hierarchy can also be queried at intermediate granularities that are not part of the main Level 1 to Level 3 sequence. Two examples are shown here. The first restricts the atlas to labels containing a commissural component, highlighting how callosal territories change across scale. The second groups labels by dominant lobar connections, providing a lobe-based view of the same atlas.

#### *Commissural-containing labels across scales*

The corpus callosum provides a useful example of how the atlas changes across scales. At Level 1, the midline is dominated by commissural connectivity, consistent with the callosal core. At Level 2, this pattern broadens laterally into association-commissural and commissural-projection territories, especially in superior frontal and parietal white matter. At Level 3 and at the full atlas level, the isthmus and splenium show the greatest internal complexity, with distinct C, C-C, C-C-C, and C-C-C-C patterns occupying reproducible topographic positions. **Figure 7** shows this progression across hierarchical levels by restricting the display to labels containing a commissural component. The simplest organization is seen near the midline, whereas more complex mixed patterns emerge as callosal pathways move laterally into surrounding deep white matter. This pattern emphasizes that the callosal midline is dominated by broad lobar commissural signatures, whereas finer-scale complexity emerges most clearly in the isthmus and splenium. Because these are lobar-based rather than gyral-based connectivity signatures, the figure is less a parcellation of classical callosal subdivisions than a map of how posterior commissural organization becomes progressively more differentiated across atlas levels.

#### *Dominant lobar connectivity views*

The atlas can also be used to group and study labels according to their dominant lobar connections. Additional lobe-based views of the atlas are provided in Supplementary Figure 1, illustrating how the same labels can be queried according to dominant endpoint territory.



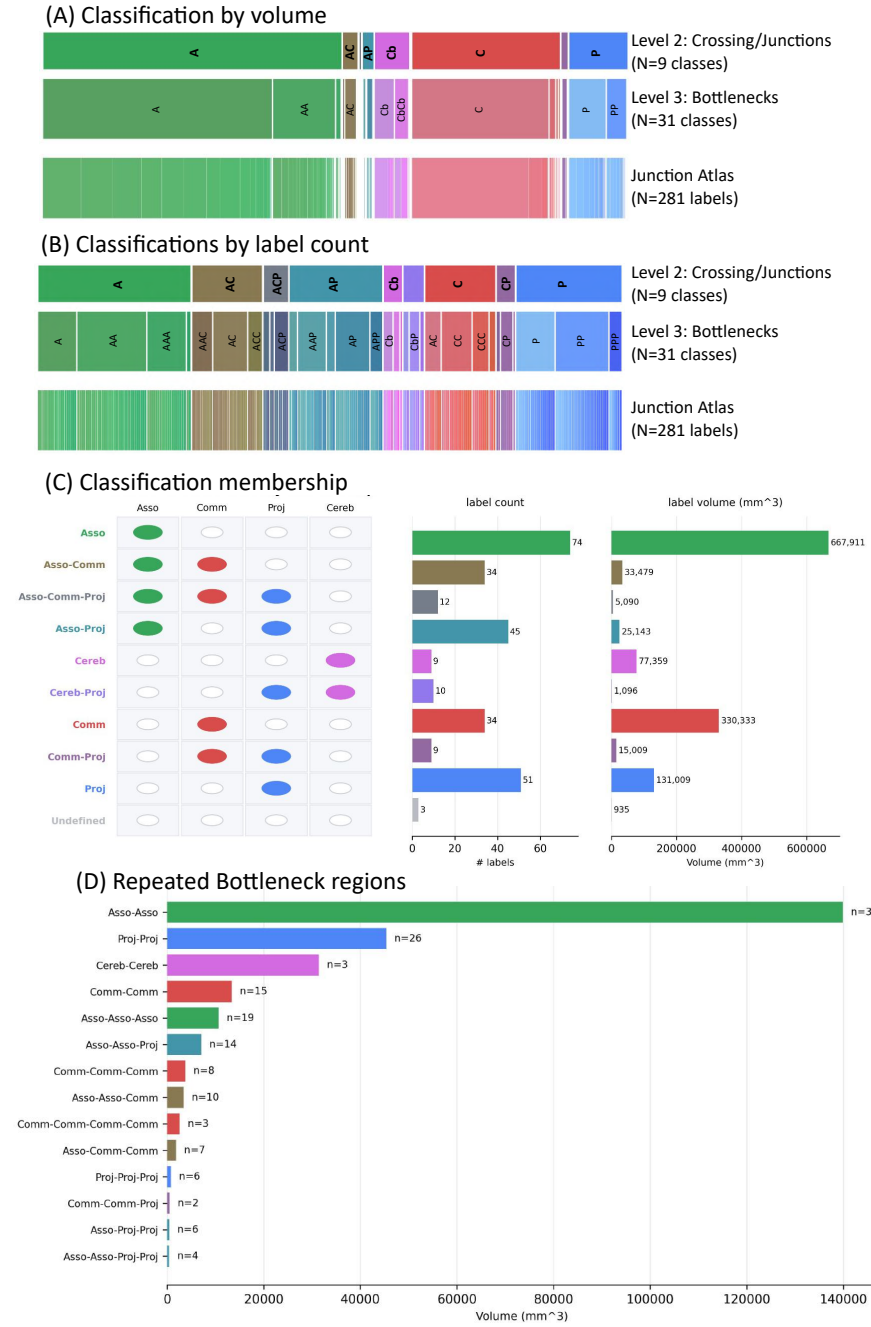
**Figure 7. Hierarchical organization of commissural-containing atlas labels.** Only labels with a commissural component are displayed, shown at the broad system level 1, the Level 2 junction level, the Level 3 overlap level, and the full atlas level in midsagittal and parasagittal slices. Midline territories are dominated by commissural connectivity, whereas lateral slices reveal association-commissural and commissural-projection transitions and progressively finer topographic differentiation within the isthmus and splenium

#### Atlas composition and reproducibility across scales

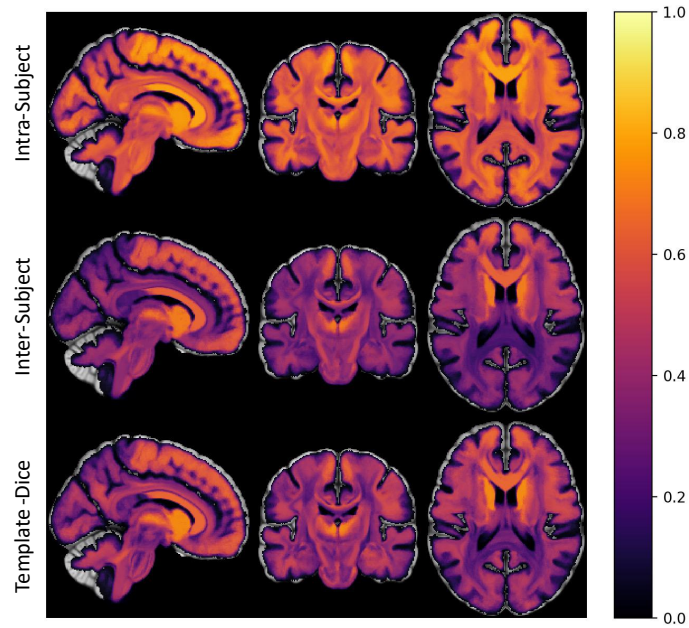
Across hierarchical levels, the atlas is dominated by association pathway patterns (**Figure 8**). By both total volume and label count (**Figure 8A, B**), association classes contribute the largest share of the atlas, followed by commissural classes, projection, then cerebellar (a subset of projection). At the Level 2 scale (**Figure 8C**), this remains true. Among mixed configurations, association-commissural and association-projection patterns are the largest and most frequent. Among repeated bottleneck regions (**Figure 8D**), repeated association patterns are especially common, with large A-A territories (N=34) forming much of the external zone, while P-P classes (N=26) are the next most prominent. This is followed by cerebellar-cerebellar (N=3, but a large overall volume) and commissural-commissural (N=15) patterns.

To examine reproducibility in space, we created a set of Dice maps from the scan-rescan HCP cohort (Figure 10). For each label, we first computed its scan-rescan Dice coefficient and then assigned that value to all voxels occupied by that label, averaging across subjects. The resulting image provides a spatial summary of the expected local reproducibility of junction labels, with higher values indicating locations where the same label is more consistently recovered across repeated scans. Using the same approach, we also generated inter-subject and subject-to-template Dice maps (**Figure 9**), which summarize where label assignments are spatially consistent across individuals and where they most closely agree with the subject-averaged atlas template. Scan-rescan agreement is generally high across much of the atlas, suggesting that broad label placement is stable within subjects even when some individual labels are small. Inter-subject (and subject-to-template) agreement was lower and more spatially variable, particularly near parietal and occipital lobes.

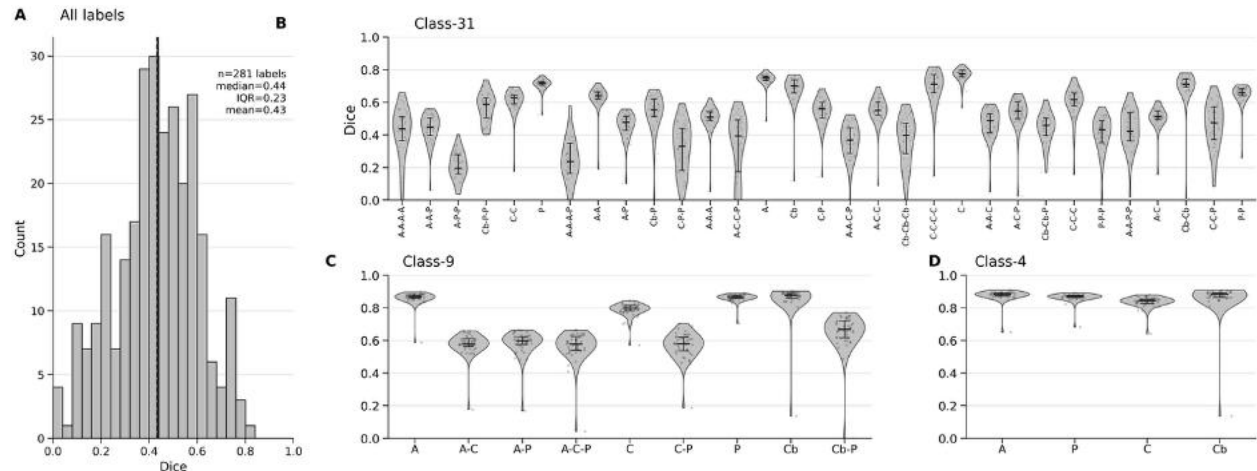
This spatial pattern is mirrored in the scan-rescan Dice distributions across hierarchical levels (**Figure 9**). At the full 281-label level, the median of the per-label median Dice values is approximately 0.44 (IQR = 0.33-0.56). While individual label scores are frequently in the low-to-intermediate range, this likely reflects their small spatial extent (mean = 4581 voxels; median = 366 voxels; mode = 4 voxels). With small regions, even slight spatial offsets can reduce Dice sharply. Reproducibility increases at broader scales. At Level 1 (Class-4), median Dice is high for all four broad systems, with association 0.883, projection 0.871, commissural 0.843, and cerebellar 0.883. At Level 2 (Class-9), the single-family categories remain highest, whereas mixed groups such as association-commissural, association-projection, commissural-projection, and association-commissural-projection are lower. Level 3 (Class-31) shows the widest spread, with relatively reproducible simple or repeated classes in some cases, but substantially lower Dice for several mixed bottleneck patterns, including association-projection-projection. Overall, the bottleneck-specific regions are spatially variable across individuals.



**Figure 8. Hierarchical composition of atlas classes across scales.** (A) Total atlas volume (mm<sup>3</sup>) summarized by class across hierarchical levels. (B) Number of individual labels contributing to each class. (C) Membership patterns across levels, showing how often different system combinations occur within the atlas. (D) Common Level 3 bottleneck-pattern classes summarized by their frequency and spatial extent (mm<sup>3</sup>).



**Figure 9. Spatial distribution of atlas reproducibility.** Maps show intra-subject, inter-subject, and template-based Dice agreement across the atlas. For each comparison, a single label-level Dice score is assigned to all voxels belonging to that label and then averaged across contributing images. Voxel intensity does not represent a voxel-wise Dice value, but the cohort-averaged reproducibility of the label occupying that location. Values closer to 1 indicate labels that are more consistently recovered in the same spatial location across the comparison, whereas values closer to 0 indicate labels with little spatial overlap. These maps capture both label reproducibility and how consistently a given label appears at the same location across subjects.



**Figure 10. Scan-rescan Dice across hierarchical levels of the atlas.** (A) Dice values for all individual labels across 41 scan-rescan pairs. Violin plots summarizing Dice distributions after grouping labels at the Class-31 (B), Class-9 (C), and Class-4 (D) levels. Central bars indicate median and quantiles. Reproducibility increases as the atlas is collapsed from fine individual labels to broader hierarchical groupings. Class-4 masks are overlapping broad unions rather than mutually exclusive categories.

## Discussion

We developed a connectivity-driven atlas of human deep white matter centered on junction areas, defined here as spatially discrete territories where multiple fiber systems occupy the same macroscopic space. Using tractography-derived connectivity signatures and joint label fusion across 105 subjects, we parcellated the white matter into 281 reproducible labels and organized them across several levels of anatomical detail. At the broadest level, the atlas recovers the major association, commissural, projection, and cerebellar systems. At progressively finer levels, it identifies inter-system junctions, dense overlap patterns, and local territories with distinct connectivity profiles. These levels provide a way to organize regions where white matter pathways merge, cross, run alongside one another, or diverge, areas that are often difficult to account for with existing regional or tract-based atlases.

### From regions and tracts to deep white matter junctions

Classical white matter anatomy has traditionally been organized around named tracts and broad anatomical regions (Bullock et al., 2022). This framework remains essential, but it does not fully describe the local territories where multiple pathways converge, overlap, or share the same deep white matter corridor. The Junction Atlas adds this missing topographic layer. Rather than replacing tract-based anatomy, it identifies reproducible junction territories within familiar regions such as the centrum semiovale, corona radiata, internal capsule, and corpus callosum. This shift from tracts to junctions is important because deep white matter complexity is not limited to regions where pathways cross at sharply different angles. The atlas also identifies territories where distinct pathway populations run in parallel, form repeated intra-system components or occupy compact bottlenecks before diverging toward separate cortical or subcortical targets.

This shift is the next intuitive step in the evolution of white matter atlases. Early voxel-based templates, including the JHU and ICBM-DTI atlases, gave standardized names to white matter locations in template space, making group studies and lesion reports more interpretable across datasets (Huang et al., 2005; S. Mori et al., 2008; Susumu Mori, Wakana, Van Zijl, & Nagae-Poetscher, 2005; Oishi et al., 2008). This remains an essential (and most common) use of atlases. However, these labels usually assign one dominant anatomical name to a voxel. In regions such as the centrum semiovale, corona radiata, and internal capsule, this can force broad, pathway-rich territories into a single regional label, even though several systems may pass through the same macroscopic volume. Tractography-based atlases shifted the focus from locations to pathways. Virtual dissection protocols and bundle atlases reconstructed long-recognized association, commissural, projection, and cerebellar pathways, producing population probability maps, streamline atlases, and automated segmentation tools for tract-specific analysis (M. Catani & Thiebaut de Schotten, 2008; Hansen et al., 2020; Hua et al., 2008; Warrington et al., 2020; Yendiki et al., 2011; Zhang et al., 2018). These resources made it possible to study named

fasciculi across subjects. They also made overlap immediately apparent, with major bundles shown to often share, cross, or converge within the same deep white matter territories.

Because of this, a single white matter location can therefore carry several descriptions at once. It may lie within a regional atlas *label*, within one or more named *tracts*, and within a larger multi-system *territory*. Recent resources have started to directly represent this layered organization. RegionConnect assigns multiple connectivity labels to each white matter voxel, allowing voxel-wise findings to be related to standardized connectional information (Qi & Arfanakis, 2021). Volz and colleagues mapped crossing-fiber pockets from local orientation structure, providing compartments in which diffusion measures could be interpreted with reduced variability and drawing attention to anatomically complex regions (Volz et al., 2018). Related multimodal approaches integrate structural connectivity with functional information, extending white matter mapping beyond structural anatomy alone (Nozais et al., 2023; Zhou et al., 2026). These resources describe overlapping connectivity, local orientation structure, or functional organization. The recurring three-dimensional territories formed by this overlap have been less directly organized as anatomical units that can be localized, compared across subjects, and interpreted across scales.

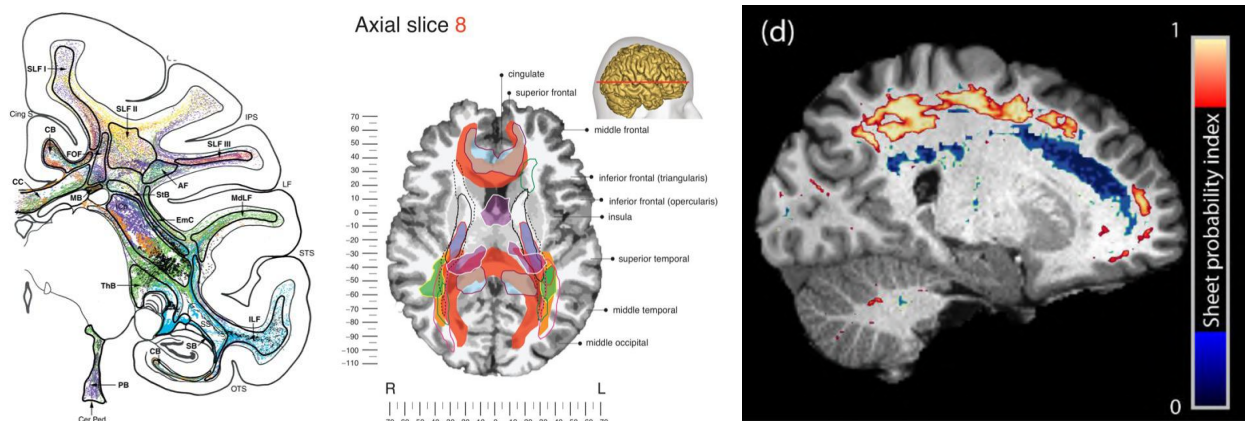
The WM Junction Atlas was designed to sit between these existing approaches. It retains the spatial specificity of voxel-based atlases, the connectional information of tractography atlases, and the layered view of multi-label resources, while grouping recurring overlap patterns into discrete junction territories. Each label can be described by its position, shape, connectivity signature, system composition, hierarchical class, and reproducibility. This atlas can be read (and used) at several scales. A highly localized finding can be assigned to one of the 281 labels. A broader effect can be summarized by Level 3 overlap patterns or Level 2 junction classes. Still broader analyses can collapse the atlas into major systems, lobar groupings, or other derived views. In practice, this means that the same coordinate can be described as a specific label, a junction pattern, a major system combination, or a lobe-level territory, depending on the biological or clinical question.

#### Junction motifs in classical and modern white matter anatomy

Several junction patterns described by the current atlas have clear precedents in classical and modern white matter anatomy (**Figure 11**). Axonal tracing studies in non-human primates (Pandya & Yeterian, 1985; Jeremy D. Schmahmann & Pandya, 2006; J. D. Schmahmann et al., 2007), Klingler-based human white matter dissections (Wysiadecki et al., 2019), and early diffusion tractography studies (M. Catani & Thiebaut de Schotten, 2008; Marco Catani & Thiebaut de Schotten, 2012) work have repeatedly described regions where distinct pathway systems converge, interweave, or share compact anatomical corridors ("Atlas cerebri humani: The Inner Structure of the Brain Demonstrated on the Basis of Macroscopical Preparations," 1956; De Benedictis et al., 2016; Fernandez-Miranda et al., 2008; Martino et al., 2011; Sarubbo et al., 2016; Vavassori et al., 2025) (**Figure 11a,b**). The WM Junction Atlas provides a standardized volumetric representation of such overlap zones, transforming anatomical configurations that were previously described qualitatively or illustrated case by case into reproducible, measurable labels.

A related geometric language has emerged around white matter *sheets* and *bottlenecks* (**Figure 11c**). Sheet-like organization has been described as lamellar, interwoven arrangements of pathways, potentially reflecting a simple biological organization for efficient wiring structures in the complex brain (Wedeen et al., 2012). Regions of high sheet probability have been reported where callosal fibers intersect with the cingulum and corticospinal tract, regions that closely align with our association-commissural and commissural-projection junction classes (Tax et al., 2016; Tax et al., 2017). Although the WM Junction Atlas does not explicitly quantify sheet-like organization, it separates these interwoven territories into discrete labeled volumes, making it possible to ask whether different sheet-like segments show distinct microstructural profiles or developmental trajectories.

The bottleneck concept has a parallel history. Mangin and colleagues (Mangin, Régis, & Frouin) described major structures such as the corpus callosum, anterior commissure, and brainstem as *shape* bottlenecks, while later tractography work localized bottleneck-like configuration in deep occipital white matter, the brainstem, and the posterior limb of the internal capsule, where multiple bundles share a dominant orientation before separating. Our atlas assigns structured junction labels to several of these locations, adding an anatomical and hierarchical description to regions that have otherwise been discussed mainly as geometric constraints or sites of tractography ambiguity.



**Figure 11. Junction motifs in classical and modern white matter anatomy.** Overlap zones mapped in the Junction atlas build on foundational observations from classic and modern modalities. This includes the convergence of tracers in non-human primates (A), spatial overlap of major named bundles extracted with diffusion tractography (B), and a blade of sheet-like structures (C). Panel A: Schmahmann and Pandya, *Fiber Pathways of the Brain*, Fig. 27-1, p. 544, Copyright © 2006 by Oxford University Press, Inc. Panel B: Catani and Thiebaut de Schotten, *Atlas of Human Brain Connections*, Ch. 6, Axial Slice 8, pp. 140–141, © Oxford University Press, 2012. Panels A-B reproduced with permission of the Licensor through PLSclear; panel C reproduced with permission from Tax et al. (2016). OUP content in panels A-B is excluded from all forms of open access license, including Creative Commons, and may not be reused without permission of Oxford University Press.

At a broader scale, the hierarchy of the atlas reveals a topological nesting that mirrors the concentric white matter zones first proposed by Theodor Meynert in the 19th century (Meynert, 1885, 1892). Our multi-level classification stratifies the deep structural core - dominated by projection and brainstem fibers - from the intermediate commissural wrap and the more superficial association layers. Recognizing this spatial layering is highly relevant for tracking structural

changes across the lifespan. White matter myelination, maturation, and eventual age-related degeneration do not occur uniformly; instead, they follow a stereotyped spatial gradient that generally progresses from deep to superficial and posterior to anterior. Because our parcellation inherently divides the white matter along these exact topological boundaries, it offers a built-in spatial scaffold to measure these lifespan trajectories. Rather than evaluating isolated tracts, researchers can track how distinct deep-to-superficial intersection tiers mature over time, potentially linking them to specific phases of cognitive development.

Finally, the prominence and complexity of association-driven junctions in the atlas are consistent with the evolutionary expansion of human cortical association networks. Recent connectomic frameworks emphasize that complex cognitive functions rely on highly variable, densely interconnected systems within the expanded human cortex (Thiebaut de Schotten and Forkel, 2022). In the present atlas, many of the most spatially intricate labels involve association pathways interacting with commissural, projection, or other association components. These intersection zones may therefore represent candidate anatomical substrates for functional integration, hemispheric specialization, and vulnerability to network-level disconnection. Because the current implementation uses relatively coarse lobar endpoint classes, the association-related complexity described here is likely conservative; finer gyral or multimodal endpoint parcellations may further subdivide these territories. Establishing standardized boundaries for these junctions provides a foundation for testing how individual differences in intersection anatomy relate to cognitive diversity, lateralization, and complex clinical disconnection profiles.

#### Rationale and applications of the WM Junction Atlas

Mapping white matter junction territories is important for both basic and clinical neuroscience. For the anatomist, the atlas provides reproducible landmarks for studying the morphological relationships among major pathway systems, including blade-like (Oishi et al., 2008), sheet-like (Galinsky & Frank, 2016; Tax et al., 2016; Tax et al., 2017; Wedeen et al., 2012), and compact bottleneck-shaped territories (Schilling et al., 2021) that are difficult to describe with tract names alone. For the neurologist, these same territories provide a framework for interpreting disconnection, since a small lesion in a junction area may affect several systems at once and produce deficits that do not map cleanly onto a single tract. For example, damage to a compact association-projection or commissural-projection junction may produce a mixed clinical profile that would be difficult to interpret using a single tract label alone. For the neurosurgeon, junctions may help estimate operative risk by identifying locations where multiple distinct functional systems share the same anatomical location.

For connectomics and functional imaging, junction areas represent the physical substrate of high edge overlap, where many long-range connections pass through a limited volume and may shape network efficiency and vulnerability. For tractography, the atlas identifies where local reconstruction is most likely to be ambiguous, providing spatial priors for algorithm development, validation, and cautious interpretation. Overall, the WM Junction atlas can be used as both a simple localization tool and a standardized description of organization that can make these territories useful for analysis and interpretation.

#### Limitations and Future directions

The WM Junction atlas is derived from diffusion MRI fiber tractography and therefore inherits the known limitations of diffusion data and streamline reconstruction. These include false negative and false positive connections (especially prevalent in the bottlenecks that motivated this atlas) as well as variability in the tracking propagation algorithms – all of which influence the connectivity signatures assigned to each voxel. This map should be interpreted as a map of recurring macroscopic connectivity *as reconstructed with tractography*. Future work should explicitly test the invariance of the atlas across acquisition quality, diffusion models, tractography algorithms, streamline filtering strategies, and processing pipelines, using benchmark-style validation frameworks analogous to Tractometer- or qBench-like approaches (Cote et al., 2013; Schilling et al., 2019).

The next challenge is building population consensus. Joint label fusion allowed subject-level signature maps to be combined into reproducible group labels, but the fusion step depends on registration, FA-guided spatial similarity, the target template, and the subject-level labels. It also favors patterns that recur across subjects. Rare junction arrangements, individual variants, and atypical anatomy may be underrepresented or displaced. This is not unique to white matter, and atypical topology in individual subjects may require individual-level classification (i.e., generating labels directly from tractography on that subject) rather than simple alignment to a group map.

Finally, the parcellation reflects choices made throughout the pipeline. The endpoint scheme was intentionally broad, based on lobe-scale and major subcortical divisions, yet it still produced 281 labels, a granularity on the same general order as modern cortical parcellations. This suggests that the current atlas is not overly sparse, but finer cortical and subcortical parcellations may reveal additional junction subdivisions. Future work should test the stability of the labels under different endpoint parcellations, tractography algorithms, streamline propagation rules, thresholding choices, registration methods, and fusion parameters.

### Availability

The WM Junction Atlas is made available through a public repository (10.5281/zenodo.21140524; [https://github.com/frheault/nf-junction\\_atlas/tree/main](https://github.com/frheault/nf-junction_atlas/tree/main)). The release includes the 281-label atlas in standard MNI space, hierarchical versions of the atlas at the major system, junction, and bottleneck levels, label lookup tables, connectivity-signature metadata, and reproducibility maps. These files will allow users to assign coordinates, lesion masks, activation clusters, microstructural measurements, or tractography-derived maps to atlas labels and to summarize findings across multiple anatomical scales. We have also provided an end-to-end pipeline to assign labels to new datasets.

## **Materials and Methods**

### Participants and Data Acquisition

Data for the primary atlas construction were obtained from the Human Connectome Project (HCP) Young Adult cohort (Glasser et al., 2013; Van Essen et al., 2012) (data is publicly available at <https://www.humanconnectome.org/study/hcp-young-adult/data-releases>). The primary population template was generated using a randomly selected subset of 105 healthy adult subjects (60F/45M; age range 22-35). To evaluate reproducibility, we utilized the separate HCP Test-Retest dataset. This consisted of 41 subjects with repeated scan sessions (session 1 and

session 2) to assess intra-subject variance, and an additional subset of 82 distinct subjects to evaluate inter-subject variability. All data were acquired on a customized Siemens 3T Connectome Skyra scanner using the standard HCP multiband diffusion MRI (dMRI) and high-resolution T1-weighted structural imaging protocols.

### Image Preprocessing and Tractography

Processing of structural and diffusion data was done using a standardized, automated Nextflow pipeline (nf-neuro). High-resolution T1-weighted images were processed using FreeSurfer (recon-all) to generate cortical and subcortical parcellations (aparc+aseg). Diffusion MRI data underwent the HCP minimal preprocessing pipeline (eddy current, motion, and susceptibility distortion corrections)(Glasser et al., 2013). Fiber Orientation Distribution Functions (fODFs) and tensor-derived metrics, including Fractional Anisotropy (FA), were computed for each subject via the TractoFlow pipeline (Theaud et al., 2020).

To enable anatomical connectivity mapping, the T1-weighted images and corresponding FreeSurfer( Fischl, 2012) parcellations were co-registered to the native dMRI space using Advanced Normalization Tools (ANTs). Finally, whole-brain probabilistic tractography (Garyfallidis et al., 2014; Renaud et al., 2026) was generated for each subject in native space using local tracking algorithms seeded from the white matter mask.

### Definition of the Nodal Framework

To create a macroscale framework of white matter connectivity, the FreeSurfer aparc+aseg parcellation was simplified into 15 broad bilateral regions: encompassing the major cortical lobes, deep subcortical nuclei, brainstem, and cerebellum. Because connections are undirected, this 15-node network yields an upper triangular connectivity matrix representing 120 possible unique region-to-region connections  $[15 \times (15 + 1) / 2]$ .

To facilitate downstream hierarchical classification, an expert neuroanatomist manually categorized each of these 120 possible connections into classical pathway systems based on their structural terminals: Association (A; intra-hemispheric cortical), Commissural (C; inter-hemispheric cortical), Projection (P; cortex to subcortex/brainstem/spinal cord), Cerebellar (Cb), and Brainstem pathways.

### Voxel-Wise Connectivity Signatures

For every subject, the whole-brain tractogram was used to generate a voxel-wise connectivity matrix. At each white matter voxel, the number of streamlines belonging to each of the 120 region-to-region connections was counted. This 120-element vector was normalized (summing to 1) to represent a continuous probability distribution of the pathway systems occupying that specific voxel.

To isolate dominant reconstructed pathways and reduce the influence of low-density streamline contributions, the normalized vector was binarized using a 10% threshold. Endpoint pairs contributing less than 10% of the streamlines traversing a voxel were set to 0, while endpoint pairs exceeding this threshold were set to 1. This produced a discrete 120-element binary connectivity signature for each white matter voxel, defining which major systems traverse that macroscopic volume.

This 10% threshold was established empirically as a compromise between suppressing low-density tractography contributions and preserving multi-system overlap patterns. A lower threshold (e.g., 5%), retained many low-contribution endpoint pairs and produced many rare combinatorial signatures. A higher threshold (e.g., 20%), retained only the strongest reconstructed pathways and removed much of the junction structure that the atlas was intended to capture.

We used exact binary signatures rather than clustering signatures in the 120-dimensional vector space. Although distances such as Hamming or Jaccard distance could be computed, the vector entries correspond to endpoint-pair identities rather than ordered spatial samples, and nearby vector positions do not have a natural anatomical relationship. Exact matching therefore avoided imposing an arbitrary distance metric and preserved a direct interpretation of each label as a specific combination of reconstructed endpoint pairs.

### Subject-Level Extraction and Group-Level Label Fusion

To construct the population atlas, all subject-level signature maps were generated in standard MNI space after transforming the tractograms, endpoint parcellations, and white matter masks into the same template space (MNI152 NLIN 6<sup>th</sup> generation)(Grabner et al., 2006). Before filtering, each subject contained thousands of unique binary connectivity signatures (~8,000 per individual; ~145,000 unique signatures across the entire population). Most of these were infrequent, sparse, or isolated voxels, reflecting noise or minor anatomical variations. We retained only those unique signatures that occupied a cumulative total of at least 10,000 voxels across the entire cohort. This threshold effectively distilled the 145,000 theoretical signatures down to a highly conserved candidate set of 396 spatially distinct junction patterns.

These maps were subsequently aggregated into a single, cohesive atlas using the ANTs Joint Label Fusion (JLF) algorithm (antsJointLabelFusion.sh)(Wang et al., 2013). JLF utilized the subjects' aligned Fractional Anisotropy (FA) maps as an intensity-driven similarity metric. The algorithm spatially weighted the label voting based on local FA similarities between the individual subjects and the MNI FA target. Following JLF, the pipeline yielded 281 highly reproducible, distinct junction labels spanning the deep white matter.

### Reproducibility Analysis

To validate the spatial stability and biological reliability of the junction parcellation, spatial overlap was quantified using the Dice similarity coefficient across three distinct paradigms:

- Intra-subject (scan-rescan) reproducibility: The exact pipeline was run independently on session 1 and session 2 data for 41 HCP test-retest subjects to quantify variance introduced by acquisition and stochastic tractography.
- Inter-subject variability: To assess population variance in junction topology, pipeline outputs from the first session of the 41 participants were used to evaluate 82 random subject pairings.
- Template-to-subject agreement: The 41 individual session 1 maps were compared directly against the final JLF-derived population template to evaluate how accurately the group atlas represents an individual's native architecture.

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